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Aryabhatta Maths Olympiad (AMO)

Level 1 — Practice Paper 2

CLASS 10

Questions	45 multiple-choice questions
Marks	100
Time	60 minutes
Section I	40 questions × 2 marks = 80 marks
Section II	5 Star Questions × 4 marks = 20 marks
Negative marking	No negative marking
Calculator	Not allowed
Chapters	Number System · Fractions, Decimals & Percentages · Algebra · Geometry · Mensuration · Trigonometry · Data Handling · Applied Mathematics · Mathematical Reasoning

Instructions to candidates

- This paper has 45 multiple-choice questions. Each question has four answer choices, of which exactly one is correct.
- Section I (Q1–Q40) carries 2 marks per question. Section II (Q41–Q45) carries 4 marks per question.
- There is no negative marking. Unanswered questions score zero.
- Calculators are not permitted. Rough work may be done on a separate sheet.
- Mark your answers on the answer sheet at the end of the paper.
- All figures are not necessarily drawn to scale unless stated.

For parents and teachers

Print or view this paper in colour. Pictures are part of the questions.

Parts follow the AMO syllabus chapters. Chapter 2 is not part of the Class 10 syllabus, so the chapter numbers shown in the Part headings have gaps.

The answer sheet and the answer key are at the end. Remove only the answer key before the student begins.

Name:

School:

Roll No.:

SECTION I 40 questions × 2 marks each**Part 1 — Number System (Syllabus Chapter 1)**

Q1. Two numbers are written as products of primes: $a = 2^4 \times 3 \times 5^2$ and $b = 2^2 \times 3^3 \times 7$. Which of these is the HCF of a and b ?

- (A) $2^4 \times 3^3$ (B) $2^2 \times 3 \times 5^2 \times 7$
 (C) $2^4 \times 3^3 \times 5^2 \times 7$ (D) $2^2 \times 3$

Q2. For which of these values of k does $\frac{k}{150}$ have a terminating decimal expansion?

- (A) 21 (B) 35 (C) 40 (D) 25

Q3. Euclid's algorithm is used to find the HCF of two numbers a and b , where $a > b$. The steps are $a = 3 \times b + r$, then $b = 2 \times r + s$, then $r = 4 \times s + 0$. The HCF of a and b is 7. What is the value of a ?

- (A) 189 (B) 217 (C) 266 (D) 63

Part 2 — Fractions, Decimals & Percentages (Syllabus Chapter 3)

Q4. A machine bought for ₹80,000 loses value at 10% per year. The loss is worked out every half-year on the value at that time. What is the value of the machine after one year?

- (A) ₹64,800 (B) ₹72,000 (C) ₹72,200 (D) ₹76,000

Q5. Jai and Gita run a toy shop for one year. The profit is shared in the ratio of capital × time. Which of these plans gives them equal shares of the profit?

- (A) Jai ₹35,000 for 12 months; Gita ₹35,000 for 9 months
 (B) Jai ₹25,000 for 12 months; Gita ₹35,000 for 8 months
 (C) Jai ₹45,000 for 8 months; Gita ₹36,000 for 10 months
 (D) Jai ₹60,000 for 6 months; Gita ₹40,000 for 8 months

Q6. Pavan, Lena and Vijay start a business. Their capitals are in the ratio 2 : 3 : 5. They keep their money in the business for times in the ratio 5 : 4 : 2. In what ratio should the profit be shared?

- (A) 5 : 6 : 5 (B) 1 : 1 : 1
 (C) 2 : 3 : 5 (D) 5 : 4 : 2

Q7. Atharv and Nitya run a juice stall for a year. Atharv puts in ₹40,000 at the start, and after 4 months he adds ₹30,000 more. Nitya puts in ₹50,000 for the whole year. The year's profit is ₹15,840, shared in the ratio of capital × time. What is Atharv's share?

- (A) ₹7,200 (B) ₹7,040 (C) ₹9,240 (D) ₹8,640

Q8. ₹60,000 is deposited for one year. Which of these schemes gives the greatest amount at the end of the year?

- (A) 8% per year for the first half-year and 12% per year for the second half-year, compounded half-yearly
- (B) 10% per year, compounded half-yearly
- (C) 8% per year, compounded half-yearly
- (D) 10% per year, compounded yearly

Part 3 — Algebra (Syllabus Chapter 4)

Q9. Which of these quadratic equations has two equal roots?

- (A) $9x^2 + 30x - 25 = 0$
- (B) $x^2 - 6x + 18 = 0$
- (C) $9x^2 - 30x + 25 = 0$
- (D) $4x^2 - 8x + 16 = 0$

Q10. Which of these numbers is a term of the arithmetic progression $-4, 3, 10, 17, \dots$?

- (A) 188
- (B) 196
- (C) 200
- (D) 192

Q11. At a fruit market, 1 kg of apples, 1 kg of bananas and 1 kg of grapes together cost ₹310. Apples and bananas, 1 kg each, cost ₹190. Bananas and grapes, 1 kg each, cost ₹170. What is the price of 1 kg of bananas?

- (A) ₹50
- (B) ₹140
- (C) ₹120
- (D) ₹85

Q12. The numbers 2 and -3 are zeroes of the polynomial $x^3 + ax^2 + bx - 30$. What is its third zero?

- (A) 5
- (B) -5
- (C) 6
- (D) 1

Q13. The desks in an examination hall are set out in equal rows. Each row has 4 more desks than the number of rows. There are 117 desks in all. How many rows are there?

- (A) 11
- (B) 13
- (C) 10
- (D) 9

Q14. An arithmetic progression has first term 48. The sum of its first 10 terms is equal to the sum of its first 15 terms. What is the sum of its first 25 terms?

- (A) 300
- (B) 2,400
- (C) 0
- (D) 1,200

Part 4 — Geometry (Syllabus Chapter 5)

Q15. Which of these points is at a distance of 13 units from the point $(2, -1)$?

- (A) $(11, 3)$
- (B) $(7, 11)$
- (C) $(-6, 4)$
- (D) $(7, 13)$

Q16. A triangle has sides 4 cm, 6 cm and 8 cm. A second triangle is similar to it. Which of these could be the sides of the second triangle?

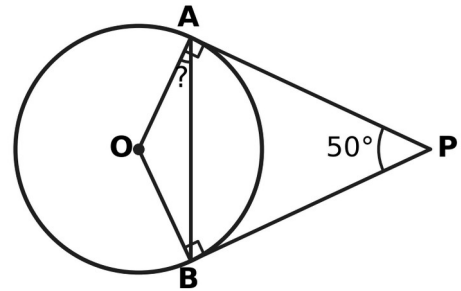
- (A) 6 cm, 9 cm and 12 cm (B) 5 cm, 7 cm and 9 cm
(C) 6 cm, 8 cm and 10 cm (D) 8 cm, 12 cm and 14 cm

Q17. The point $M(1, 2)$ lies on the segment AB , where A is $(-1, 4)$ and $AM : MB = 1 : 2$. What are the coordinates of B ?

- (A) $(3, 0)$ (B) $(7, -4)$ (C) $(2, 1)$ (D) $(5, -2)$

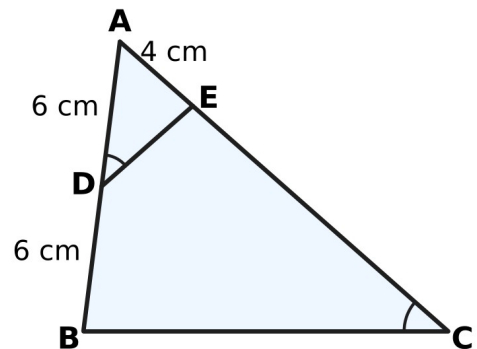
Q18. In the figure, PA and PB are tangents from P to the circle with centre O , and $\angle APB = 50^\circ$. The chord AB is drawn. What is the measure of $\angle OAB$?

- (A) 65° (B) 40° (C) 25° (D) 130°



Q19. In the figure, D lies on AB and E lies on AC , and $\angle ADE = \angle ACB$ (marked). $AD = 6$ cm, $DB = 6$ cm and $AE = 4$ cm. What is the length of AC ?

- (A) 18 cm (B) 13 cm (C) 8 cm (D) 12 cm



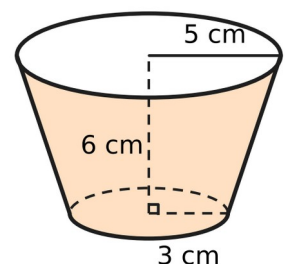
Part 5 — Mensuration (Syllabus Chapter 6)

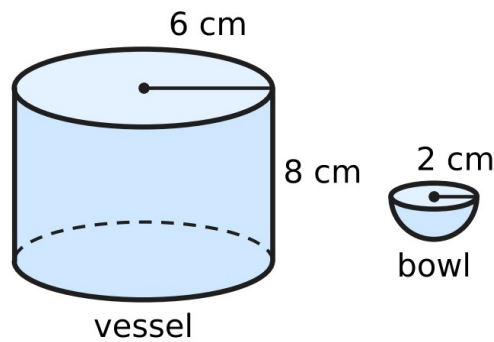
Q20. Which of these sectors of a circle has the largest area?

- (A) radius 28 cm, angle 20° (B) radius 21 cm, angle 45°
(C) radius 7 cm, angle 300° (D) radius 14 cm, angle 90°

Q21. A paper cup used at a picnic is shaped like a frustum of a cone. Its top radius is 5 cm, its bottom radius is 3 cm and its height is 6 cm, as shown. How much water can it hold when full? (Use $\pi = \frac{22}{7}$.)

- (A) 462 cm^3 (B) 308 cm^3 (C) 924 cm^3 (D) 616 cm^3



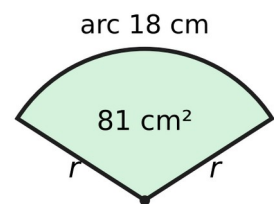


Q22. A cylindrical vessel of radius 6 cm and height 8 cm is full of water. All the water is poured into hemispherical bowls of radius 2 cm, shown in the figure, and each bowl is filled to the brim. How many bowls are filled?

- (A) 27 (B) 36 (C) 108 (D) 54

Q23. A sector of a circle has an arc of length 18 cm and an area of 81 cm^2 , as shown. What is the radius r of the circle?

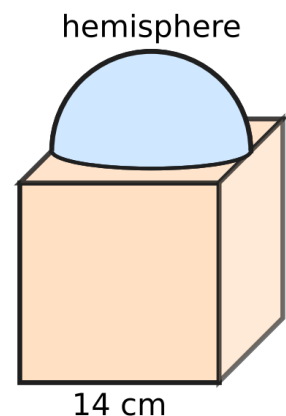
- (A) 9 cm (B) 12 cm (C) 4.5 cm (D) 18 cm



Q24. A solid is made of a cube of edge 14 cm with a hemisphere of radius 7 cm fixed on its top face, as shown. What is the total surface area of the solid?

(Use $\pi = \frac{22}{7}$.)

- (A) $1,022 \text{ cm}^2$ (B) $1,484 \text{ cm}^2$
(C) $1,330 \text{ cm}^2$ (D) $1,176 \text{ cm}^2$



Part 6 — Trigonometry & Heights and Distances (Syllabus Chapter 7)

Q25. Which of these has the greatest value?

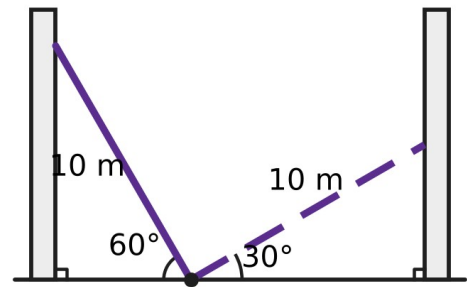
- (A) $\sin 60^\circ + \cos 60^\circ$ (B) $\sec 45^\circ$
(C) $\tan 60^\circ$ (D) $\operatorname{cosec} 30^\circ$

Q26. Which of these is equal to $\sin^4 \theta - \cos^4 \theta$ for every angle θ ?

- (A) $2 \sin^2 \theta - 1$ (B) $(\sin \theta - \cos \theta)^2$
(C) $1 - 2 \sin^2 \theta$ (D) 1

Q27. A ladder 10 m long stands with its bottom end at one point of a level lane between two vertical walls. Leaning on one wall, it makes an angle of 60° with the ground. Turned about its bottom end to lean on the other wall, it makes 30° with the ground, as shown. How far apart are the two walls?

- (A) 20 m
(B) $5(\sqrt{3}-1)$ m
(C) $5(\sqrt{3}+1)$ m
(D) $5\sqrt{3}$ m



Part 7 — Data Handling, Statistics & Probability (Syllabus Chapter 8)

Lengths of the fish in an aquarium

Length (cm)	2-6	6-10	10-14	14-18	18-22
Number of fish	3	7	6	3	1

Q28. The table shows the lengths of the 20 fish in an aquarium. What is the mean length of a fish, using the class marks?

- (A) 8 cm
(B) 10.4 cm
(C) 8.4 cm
(D) 12 cm

Q29. Bowl X has 1 red bead and 3 white beads. Bowl Y has 1 red bead and 4 white beads. One bead is taken at random from each bowl. What is the probability that both beads are white?

- (A) $\frac{2}{5}$
(B) $\frac{3}{5}$
(C) $\frac{7}{9}$
(D) $\frac{1}{20}$

Q30. A fair coin is tossed and a fair die is thrown at the same time. Which of these events has the greatest probability?

- (A) a head, or a number from 3 to 6
(B) a number greater than 2
(C) a head, or a number greater than 3
(D) a tail and a number less than 6

Part 8 — Applied Mathematics (Syllabus Chapter 9)

Q31. A shoe rack is marked ₹2,400. Which of these offers gives the lowest price?

- (A) a single discount of 30%
(B) 20% off, then a further 15% off
(C) 10% off, then a further 25% off
(D) 15% off, then a further 15% off

Q32. Akash paints twice as fast as Rekha. Working together, they paint a fence in 6 days. How many days would Rekha take to paint the fence alone?

- (A) 3 days
(B) 12 days
(C) 9 days
(D) 18 days

Q33. A bill for goods sold within a state shows SGST of ₹63, charged at 9%. CGST is charged at the same rate. What is the price of the goods before GST?

- (A) ₹1,400 (B) ₹826 (C) ₹700 (D) ₹350

Q34. A tap fills an empty water tank in 3 hours. Because of a leak at the bottom, the tank takes 3 hours 30 minutes to fill. If the tank is full and the tap is closed, how long will the leak take to empty it?

- (A) 21 hours (B) 0.5 hours
(C) 10.5 hours (D) 6.5 hours

Q35. Two towns stand 80 km apart on the bank of a river that flows at 4 km/h. Two boats, each with a speed of 16 km/h in still water, start at the same time from the two towns towards each other. How long after starting do they meet?

- (A) 2 hours (B) 5 hours
(C) 3 hours 20 minutes (D) 2 hours 30 minutes

Part 9 — Mathematical Reasoning & Problem-Solving (Syllabus Chapter 10)

Q36. A tree has 8 nests. Birds settle in them, and every bird is in some nest. What is the least number of birds that makes sure that some nest has at least 5 birds?

- (A) 32 (B) 33 (C) 40 (D) 41

Q37. How many of the whole numbers from 1 to 100 are divisible by 2 or by 5 (or by both)?

- (A) 40 (B) 60 (C) 70 (D) 50

Q38. Two players take turns to remove 1, 2, 3 or 4 matchsticks from a pile of 38. The player who removes the last matchstick wins. How many matchsticks should the first player remove at the start to be sure of winning?

- (A) 3 (B) 1 (C) 4 (D) 2

Q39. On sports day, 6 teams play a tug-of-war league. Each team plays every other team once, and every match has a winner (no draws). What is the greatest number of teams that can each win at least 4 matches?

- (A) 5 (B) 4 (C) 2 (D) 3

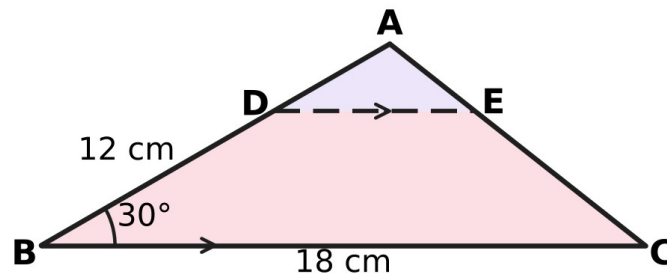
Q40. How many different triangles have sides that are whole numbers of centimetres and a perimeter of 15 cm? Triangles with the same three side lengths count as the same triangle.

- (A) 6 (B) 3 (C) 7 (D) 19

SECTION II — Star Questions 5 questions × 4 marks each

Q41. The berths in a train coach are numbered so that the numbers form an arithmetic progression. The first berth is numbered 2 and the 24th (last) berth is numbered 71. Passengers each take a different berth of this coach. What is the least number of passengers that makes sure that four of them have berth numbers with the same tens digit, where 2, 5 and 8 count as having tens digit 0? [4 marks]

- (A) 25 (B) 31 (C) 22 (D) 23



Q42. A triangular piece of cloth ABC has $AB = 12$ cm, $BC = 18$ cm and $\angle ABC = 30^\circ$, as shown. It is cut along DE, where D is on AB, E is on AC, $DE \parallel BC$ and $AD : DB = 1 : 2$. What is the area of the piece DBCE? [4 marks]

- (A) 72 cm^2 (B) 36 cm^2 (C) 48 cm^2 (D) 54 cm^2

Q43. The three sides of a right triangle, in cm, form an arithmetic progression, and its perimeter is 48 cm. A perpendicular is drawn from the vertex of the right angle to the longest side. How long is this perpendicular? [4 marks]

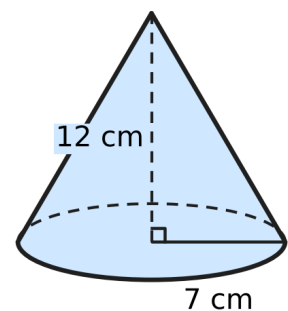
- (A) 9.6 cm (B) 7.2 cm (C) 10 cm (D) 12 cm

Q44. Mira has ₹50,000 in a bank that pays interest at 12% per year, compounded half-yearly. She wants a laptop priced ₹50,000. The shop lets her either pay now with a 10% discount, or pay the full ₹50,000 one year from now with no extra charge. Any money left unspent stays in the bank. Which way leaves her with more money one year from now, and by how much? [4 marks]

- (A) Paying later, by ₹1,180 (B) Paying later, by ₹562
(C) Paying now, by ₹562 (D) Paying later, by ₹400

Q45. A solid clay cone of radius 7 cm and height 12 cm, shown in the figure, is reshaped, with no clay lost, into a cuboid. Each edge of the cuboid is a whole number of centimetres. What is the least possible total surface area of the cuboid? (Use $\pi = \frac{22}{7}$.) [4 marks]

- (A) 221 cm^2 (B) 540 cm^2 (C) 508 cm^2 (D) 442 cm^2



Answer Sheet AMO Level 1 • Class 10 • Practice Paper 2**Name:****School:****Roll No.:***Shade or tick one circle for each question.*

- | | | |
|---------------------|---------------------|---------------------|
| 1. (A) (B) (C) (D) | 16. (A) (B) (C) (D) | 31. (A) (B) (C) (D) |
| 2. (A) (B) (C) (D) | 17. (A) (B) (C) (D) | 32. (A) (B) (C) (D) |
| 3. (A) (B) (C) (D) | 18. (A) (B) (C) (D) | 33. (A) (B) (C) (D) |
| 4. (A) (B) (C) (D) | 19. (A) (B) (C) (D) | 34. (A) (B) (C) (D) |
| 5. (A) (B) (C) (D) | 20. (A) (B) (C) (D) | 35. (A) (B) (C) (D) |
| 6. (A) (B) (C) (D) | 21. (A) (B) (C) (D) | 36. (A) (B) (C) (D) |
| 7. (A) (B) (C) (D) | 22. (A) (B) (C) (D) | 37. (A) (B) (C) (D) |
| 8. (A) (B) (C) (D) | 23. (A) (B) (C) (D) | 38. (A) (B) (C) (D) |
| 9. (A) (B) (C) (D) | 24. (A) (B) (C) (D) | 39. (A) (B) (C) (D) |
| 10. (A) (B) (C) (D) | 25. (A) (B) (C) (D) | 40. (A) (B) (C) (D) |
| 11. (A) (B) (C) (D) | 26. (A) (B) (C) (D) | 41. (A) (B) (C) (D) |
| 12. (A) (B) (C) (D) | 27. (A) (B) (C) (D) | 42. (A) (B) (C) (D) |
| 13. (A) (B) (C) (D) | 28. (A) (B) (C) (D) | 43. (A) (B) (C) (D) |
| 14. (A) (B) (C) (D) | 29. (A) (B) (C) (D) | 44. (A) (B) (C) (D) |
| 15. (A) (B) (C) (D) | 30. (A) (B) (C) (D) | 45. (A) (B) (C) (D) |

Q41–Q45 are Section II Star Questions (4 marks each).

For the parent or teacher

Section I correct: _____ / 40 × 2 = _____ marks

Section II correct: _____ / 5 × 4 = _____ marks

Correct: _____ / 45 **Score:** _____ / 100

Answer Key and Solutions AMO Level 1 • Class 10 • Practice Paper 2**Quick answers**

1 D	6 A	11 A	16 A	21 B	26 A	31 C	36 B	41 D
2 A	7 D	12 B	17 D	22 D	27 C	32 D	37 B	42 C
3 B	8 B	13 D	18 C	23 A	28 B	33 C	38 A	43 A
4 C	9 C	14 C	19 A	24 C	29 B	34 A	39 D	44 B
5 C	10 D	15 B	20 B	25 D	30 A	35 D	40 C	45 D

Solutions

Q	Part • level	Answer	Why?
1	Part 1 • Ch1 • Easy	(D) $2^2 \times 3$	The HCF takes only the common primes 2 and 3, each to its smaller power: $2^2 \times 3 = 12$.
2	Part 1 • Ch1 • Easy	(A) 21	Here $150 = 2 \times 3 \times 5^2$. The factor 3 must cancel, so k must be a multiple of 3: $\frac{21}{150} = \frac{7}{50} = 0.14$.
3	Part 1 • Ch1 • Medium	(B) 217	The HCF is the last non-zero remainder, so $s = 7$. Then $r = 4 \times 7 = 28$, $b = 2 \times 28 + 7 = 63$ and $a = 3 \times 63 + 28 = 217$.
4	Part 2 • Ch3 • Easy	(C) ₹72,200	The rate is 5% per half-year for 2 half-years: $80,000 \times 0.95 \times 0.95 = ₹72,200$.
5	Part 2 • Ch3 • Easy	(C) Jai ₹45,000 for 8 months; Gita ₹36,000 for 10 months	Capital $\times$ months: $45,000 \times 8 = 360,000$ and $36,000 \times 10 = 360,000$, which are equal.
6	Part 2 • Ch3 • Easy	(A) 5 : 6 : 5	Multiply term by term: $2 \times 5 : 3 \times 4 : 5 \times 2 = 10 : 12 : 10 = 5 : 6 : 5$.
7	Part 2 • Ch3 • Medium	(D) ₹8,640	Atharv: $40,000 \times 4 + 70,000 \times 8 = 720,000$. Nitya: $50,000 \times 12 = 600,000$. Ratio 6 : 5, so Atharv gets $15,840 \times 6/11 = ₹8,640$.
8	Part 2 • Ch3 • Medium	(B) 10% per year, compounded half-yearly	Amounts: (A) $60,000 \times 1.04 \times 1.06 = ₹66,144$; (B) $60,000 \times 1.05 \times 1.05 = ₹66,150$; (C) $60,000 \times 1.04 \times 1.04 = ₹64,896$; (D) ₹66,000. Scheme (B) gives the most.
9	Part 3 • Ch4 • Easy	(C) $9x^2 - 30x + 25 = 0$	Equal roots need $b^2 - 4ac = 0$: for $9x^2 - 30x + 25 = 0$, $900 - 4 \times 9 \times 25 = 0$. It is $(3x - 5)^2 = 0$.
10	Part 3 • Ch4 • Easy	(D) 192	A term is $-4 + 7(n - 1)$, so $192 + 4 = 196$ must be a multiple of 7: $196 = 7 \times 28$, so 192 is the 29th term.
11	Part 3 • Ch4 • Easy	(A) ₹50	Adding the two pair totals counts bananas twice: $190 + 170 = 360$, and $360 - 310 = ₹50$.
12	Part 3 • Ch4 • Medium	(B) -5	For $x^3 + ax^2 + bx + d$, the product of the zeroes is $-d = 30$. So $2 \times (-3) \times k = 30$, giving $k = -5$.
13	Part 3 • Ch4 • Medium	(D) 9	If there are x rows, $x(x + 4) = 117$, so $x^2 + 4x - 117 = 0$, which is $(x + 13)(x - 9) = 0$. So $x = 9$.
14	Part 3 • Ch4 • Medium	(C) 0	$S_{10} = S_{15}$ gives $5(96 + 9d) = 7.5(96 + 14d)$, so $d = -4$. Then $S_{25} = 12.5 \times (96 + 24 \times (-4)) = 0$.
15	Part 4 • Ch5 • Easy	(B) (7, 11)	From (2, -1) to (7, 11): $\sqrt{5^2 + 12^2} = \sqrt{169} = 13$.

Q	Part · level	Answer	Why?
16	Part 4 · Ch5 · Easy	(A) 6 cm, 9 cm and 12 cm	The ratios $6/4 = 9/6 = 12/8 = 1.5$ are equal, so the triangles are similar (SSS similarity).
17	Part 4 · Ch5 · Easy	(D) (5, -2)	By the section formula, $M = (2A + B) \div 3$, so $B = 3M - 2A = (3 + 2, 6 - 8) = (5, -2)$.
18	Part 4 · Ch5 · Medium	(C) 25°	$PA = PB$, so $\angle PAB = (180^\circ - 50^\circ) \div 2 = 65^\circ$. $OA \perp PA$, so $\angle OAB = 90^\circ - 65^\circ = 25^\circ$.
19	Part 4 · Ch5 · Medium	(A) 18 cm	Triangles ADE and ACB share $\angle A$ and have $\angle ADE = \angle ACB$, so they are similar: $AD/AC = AE/AB$, so $AC = 6 \times 12 \div 4 = 18$ cm.
20	Part 5 · Ch6 · Easy	(B) radius 21 cm, angle 45°	The area is proportional to $r^2 \times \text{angle}$: $441 \times 45 = 19,845$, more than $196 \times 90 = 17,640$, $784 \times 20 = 15,680$ and $49 \times 300 = 14,700$.
21	Part 5 · Ch6 · Easy	(B) 308 cm^3	$V = \frac{1}{3}\pi h(R^2 + r^2 + Rr) = \frac{1}{3} \times 22/7 \times 6 \times (25 + 9 + 15) = 44/7 \times 49 = 308 \text{ cm}^3$.
22	Part 5 · Ch6 · Easy	(D) 54	Water: $\pi \times 36 \times 8 = 288\pi \text{ cm}^3$. One bowl: $\frac{3}{8}\pi \times 8 = 16\pi/3 \text{ cm}^3$. $288\pi \div 16\pi/3 = 54$ bowls.
23	Part 5 · Ch6 · Medium	(A) 9 cm	Area of a sector = $\frac{1}{2} \times r \times (\text{arc length})$, because $\theta/360 \times \pi r^2 = \frac{1}{2} \times r \times (\theta/360 \times 2\pi r)$. So $81 = \frac{1}{2} \times r \times 18$ and $r = 9$ cm.
24	Part 5 · Ch6 · Medium	(C) $1,330 \text{ cm}^2$	The cube gives $6 \times 196 = 1,176$. Subtract the hidden circle 154 and add the curved surface $2 \times 154 = 308$: $1,176 - 154 + 308 = 1,330 \text{ cm}^2$.
25	Part 6 · Ch7 · Easy	(D) cosec 30°	$\text{cosec } 30^\circ = 1 \div \sin 30^\circ = 2$, while $\tan 60^\circ \approx 1.73$, $\sec 45^\circ \approx 1.41$ and $\sin 60^\circ + \cos 60^\circ \approx 1.37$.
26	Part 6 · Ch7 · Easy	(A) $2 \sin^2 \theta - 1$	$\sin^4 \theta - \cos^4 \theta = (\sin^2 \theta + \cos^2 \theta)(\sin^2 \theta - \cos^2 \theta) = \sin^2 \theta - (1 - \sin^2 \theta) = 2\sin^2 \theta - 1$.
27	Part 6 · Ch7 · Medium	(C) $5(\sqrt{3} + 1) \text{ m}$	Distances from the bottom end: $10 \times \cos 60^\circ = 5 \text{ m}$ and $10 \times \cos 30^\circ = 5\sqrt{3} \text{ m}$. Width = $5 + 5\sqrt{3} = 5(\sqrt{3} + 1) \text{ m}$.
28	Part 7 · Ch8 · Easy	(B) 10.4 cm	$\Sigma fx = 3 \times 4 + 7 \times 8 + 6 \times 12 + 3 \times 16 + 1 \times 20 = 208$, and $208 \div 20 = 10.4 \text{ cm}$.
29	Part 7 · Ch8 · Easy	(B) $\frac{3}{5}$	$P(\text{white from X}) = 3/4$ and $P(\text{white from Y}) = 4/5$. Both: $3/4 \times 4/5 = 3/5$.
30	Part 7 · Ch8 · Medium	(A) a head, or a number from 3 to 6	There are 12 equally likely outcomes. Only a tail with a 1 or 2 fails "a head, or a number from 3 to 6", so it has $10/12 = 5/6$, more than $3/4$, $2/3$ and $5/12$.
31	Part 8 · Ch9 · Easy	(C) 10% off, then a further 25% off	Prices: $2,400 \times 0.9 \times 0.75 = ₹1,620$; $\times 0.8 \times 0.85 = ₹1,632$; $\times 0.7 = ₹1,680$; $\times 0.85 \times 0.85 = ₹1,734$.
32	Part 8 · Ch9 · Easy	(D) 18 days	Rekha does 1 part and Akash 2 parts a day, so together they do 3 times Rekha's rate. Rekha alone takes $3 \times 6 = 18$ days.
33	Part 8 · Ch9 · Easy	(C) ₹700	₹63 is 9% of the price, so the price is $63 \times 100 \div 9 = ₹700$.
34	Part 8 · Ch9 · Medium	(A) 21 hours	Per hour: tap $1/3$, tap with leak $2/7$. Leak = $1/3 - 2/7 = 1/21$ of the tank per hour, so 21 hours.
35	Part 8 · Ch9 · Medium	(D) 2 hours 30 minutes	They close the gap at $(16 + 4) + (16 - 4) = 32 \text{ km/h}$: the stream helps one boat and slows the other equally. $80 \div 32 = 2.5$ hours, which is 2 hours 30 minutes.
36	Part 9 · Ch10 · Easy	(B) 33	With 32 birds, each nest can hold 4, so 32 is not enough. With 33 birds, 4 or fewer in each nest would hold at most 32, so some nest has at least 5.

Q	Part · level	Answer	Why?
37	Part 9 · Ch10 · Easy	(B) 60	There are 50 multiples of 2 and 20 multiples of 5, but the 10 multiples of 10 are counted twice: $50 + 20 - 10 = 60$.
38	Part 9 · Ch10 · Easy	(A) 3	Taking 3 leaves 35, a multiple of 5. After that, whatever the other player takes (1 to 4), the first player takes enough to make 5 in that round, so the pile stays a multiple of 5 until the first player takes the last matchstick.
39	Part 9 · Ch10 · Medium	(D) 3	There are 15 matches, so 15 wins in all. Four teams with 4 wins would need 16 wins. Three is possible: three teams beat the other three and beat each other in a cycle, $3 + 1 = 4$ wins each.
40	Part 9 · Ch10 · Medium	(C) 7	The longest side must be less than half of 15, so it is 5, 6 or 7: (5, 5, 5), (4, 5, 6), (3, 6, 6), (1, 7, 7), (2, 6, 7), (3, 5, 7), (4, 4, 7).
41	Section II · Ch4 + Ch10 · Hard	(D) 23	$d = (71 - 2) \div 23 = 3$, so the berths are 2, 5, 8, 11, ..., 71. Tens digits 0 to 6 have 3, 3, 4, 3, 3, 4, 3 berths and 7 has only 71. Taking at most 3 from each group gives $3 \times 7 + 1 = 22$ passengers with no four alike, so 23 make sure of it.
42	Section II · Ch5 + Ch7 · Hard	(C) 48 cm^2	The height of A above BC is $12 \times \sin 30^\circ = 6 \text{ cm}$, so ABC has area 54 cm^2 . By BPT, $DE = 18 \div 3 = 6 \text{ cm}$ and D is 4 cm above BC. $DBCE = \frac{1}{2} \times (6 + 18) \times 4 = 48 \text{ cm}^2$.
43	Section II · Ch4 + Ch5 · Hard	(A) 9.6 cm	Sides $a - d$, a , $a + d$: $3a = 48$, so $a = 16$. $(16 - d)^2 + 16^2 = (16 + d)^2$ gives $d = 4$: sides 12, 16, 20. The perpendicular is $12 \times 16 \div 20 = 9.6 \text{ cm}$.
44	Section II · Ch3 + Ch9 · Hard	(B) Paying later, by ₹562	Paying now: the ₹5,000 saved grows to $5,000 \times 1.06 \times 1.06 = ₹5,618$. Paying later: ₹50,000 grows to ₹56,180, and ₹6,180 is left after paying. Paying later is ahead by $6,180 - 5,618 = ₹562$.
45	Section II · Ch6 + Ch10 · Hard	(D) 442 cm^2	Volume $= \frac{1}{3} \times 22/7 \times 49 \times 12 = 616 = 2^3 \times 7 \times 11$. The surface is least when the edges are as close as possible: $7 \times 8 \times 11$, with area $2(56 + 88 + 77) = 442 \text{ cm}^2$. Checking every way of writing 616 as a product of three whole numbers confirms this; the next best, $4 \times 11 \times 14$, gives 508 cm^2 .